AI Weather and Climate Model Intercomparison Croject (AIMIP)

Systematic evaluations of AI weather and climate models

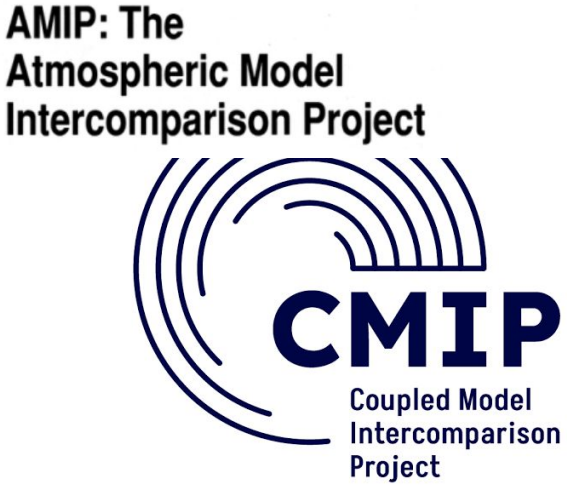
Presented by Anna Dodson

What is a MIP?

AMIP: Atmospheric Model Intercomparison Project

CMIP: Climate Model Intercomparison Project

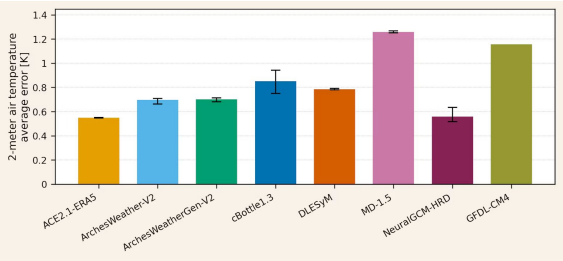
- Drives IPCC recommendations



Broadly, ***we need centralized tooling for intercomparisons of models in weather and climate***, particularly since there is no ground-truth understanding of the climate future.



What's different about AIMIP?



- We have a known baseline, CMIP class of models.
- Goal: improve trust and explainability of AI weather and climate models (AIWCMs)
- Common framework for evaluating AIWCMs + identifying differences between models

We need tooling for intercomparisons of AI and traditional models in weather and climate with each other.

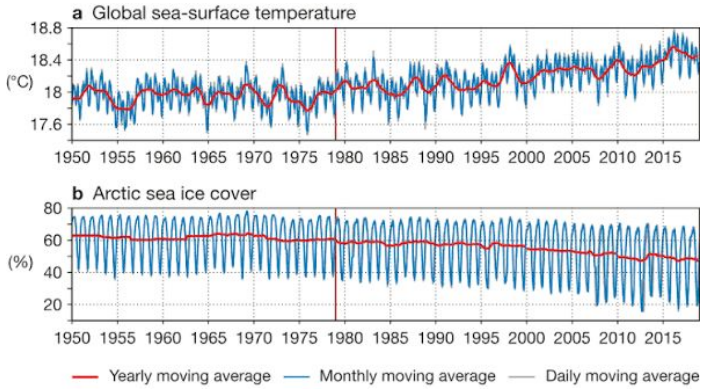
AIMIP Phase 1

Model participants simulate the atmosphere given specified historical sea surface temperatures over 1979-2024 (ERA5). Evaluate on CMIP standards:

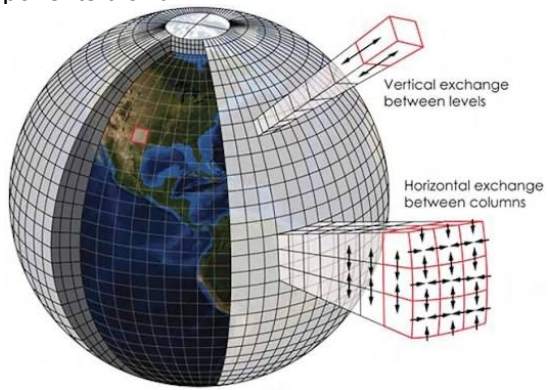
- Biases
- Trends
- Response to El Niño-related sea surface temperature anomalies,
- Temporal variability
- Out-of-sample generalization

I will go in depth into each of these in a minute.

Standardized Data Flow



- Forcings
  - Monthly SST and SIC
    - +2 K and +4 K simulations
  - No radiative forcing
  - Land may be handled by the model, but is not required
- Inputs
  - surface pressure, surface temperature, t2m, q2m, u10, v10
  - Temperature T, specific humidity q, and the eastward and northward wind components u and v
    - 1000, 850, 700, 500, 250, 100, and 50 hPa levels
- Outputs
  - all inputs + 500 hPa geopotential height



Models

All are 5-member ensembles that start on Oct. 1, 1978 and run through Dec. 31, 2024.

Table 1. AIMIP Phase 1 model submissions.

Organization	Model name	References	Code	Temporal frequency[a]	Horizontal grid	Vertical grid
Ai2	ACE2.1-ERA5	Watt-Meyer et al. (2025)	Henn et al. (2026b)	daily, monthly	$1^\circ \times 1^\circ$	13 pressure levels ('gr'), regridded from native 8 model layers via ML; native layers also submitted ('gn')
ArchesWeather	ArchesWeather	Couairon et al. (2026b); Singh et al. (2026)	Couairon et al. (2026a)	daily, monthly	$1^\circ \times 1^\circ$	7 pressure levels
ArchesWeather	ArchesWeatherGen	Couairon et al. (2026b); Singh et al. (2026)	Couairon et al. (2026a)	daily, monthly	$1^\circ \times 1^\circ$	7 pressure levels
NVIDIA	cBottle-1.3	Brenowitz et al. (2025)	Manshausen et al. (2026)	daily, monthly	HEALPix order 6 ($\sim 0.9^\circ$)	8 pressure levels
University of Washington and NVIDIA	DLESyM	Cresswell-Clay et al. (2025)	Cresswell-Clay (2026)	daily, monthly	HEALPix order 6 ($\sim 0.9^\circ$)	Some surface, 850 hPa, and 500 hPa variables
University of Maryland (PARETO group)	MD-1.5 v0.9	Hall and Molina (2026b)	Hall and Molina (2026a)	monthly	$1.5^\circ \times 1.5^\circ$ native; $1^\circ \times 1^\circ$ submitted ('gr')	7 pressure levels
Google Research	NeuralGCM	Kochkov et al. (2024); Yuval et al. (2026)	Kochkov et al. (2026)	daily, monthly	$2.8^\circ \times 2.8^\circ$	32 sigma levels native; 7 pressure levels submitted
Google Research	NeuralGCM-HRD	Kochkov et al. (2024); Yuval et al. (2026)	Kochkov et al. (2026)	daily, monthly	$1^\circ \times 1^\circ$	32 sigma levels native; 7 pressure levels submitted
NOAA GFDL	CM4 (AM4 component)	Zhao et al. (2018); Held et al. (2019)	N/A[b]	daily, monthly	C96 cubed sphere native; $\sim 1^\circ \times \sim 1^\circ$ evaluated here	33 sigma levels native; 7 pressure levels evaluated here

[a] Monthly indicates monthly average output from 1 Oct. 1978 to 31 Dec. 2024; daily indicates daily average output over (1) 1 Oct. 1978 to 31 Dec. 1979 and (2) 1 Jan. 2024 to 31 Dec. 2024.

[b] Previously published CMIP6 output used; see Code and data availability.

Table 1, Henn et al. (2026), AIMIP Phase 1: systematic evaluations of AI weather and climate models, arXiv:2605.06944.

Questions so far?

E1: Train-period and test-period time-mean bias pattern

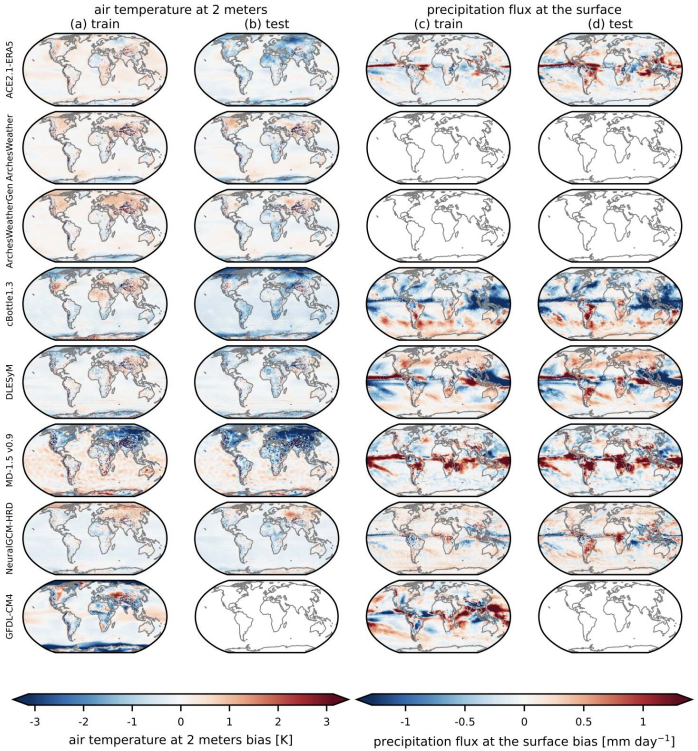
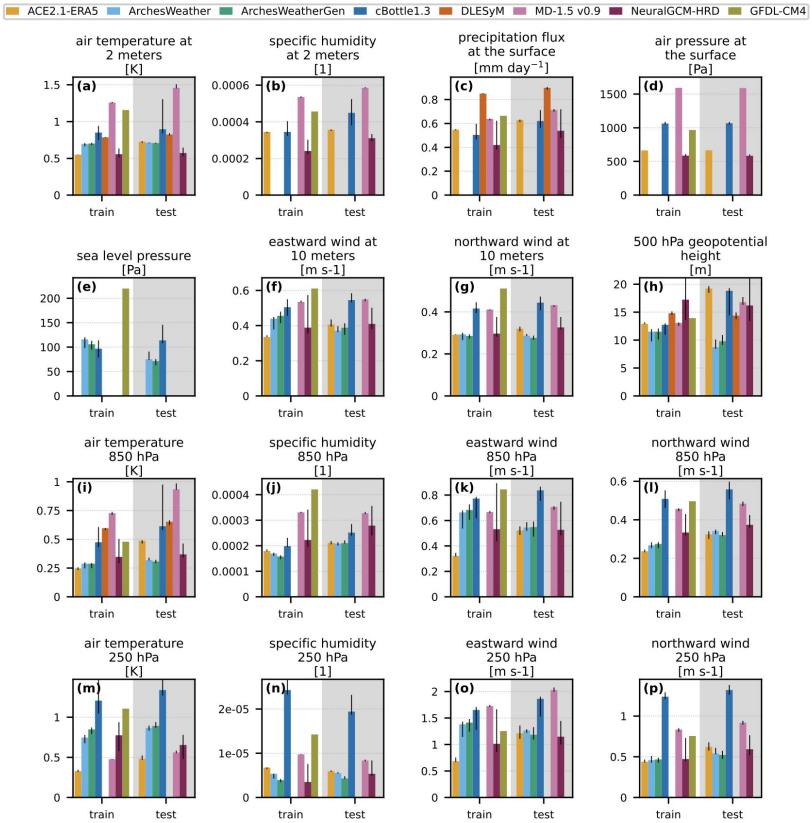


Figure 1. Biases at 1◦ resolution versus ERA5, for the AIWCMs and a CMIP6 model (GFDL-CM4, bottom row). (a), (b): 2-meter air temperature biases over the training (1979-2014) and test (2015-2024) periods, respectively. GFDL-CM4 data end in 2014 and so are only available over the training period. (c), (d): surface precipitation biases over the same periods, for models that included surface precipitation outputs (ArchesWeather and ArchesWeatherGen did not).

Fig. 1, Henn et al. (2026), AIMIP Phase 1: systematic evaluations of AI weather and climate models, arXiv:2605.06944.

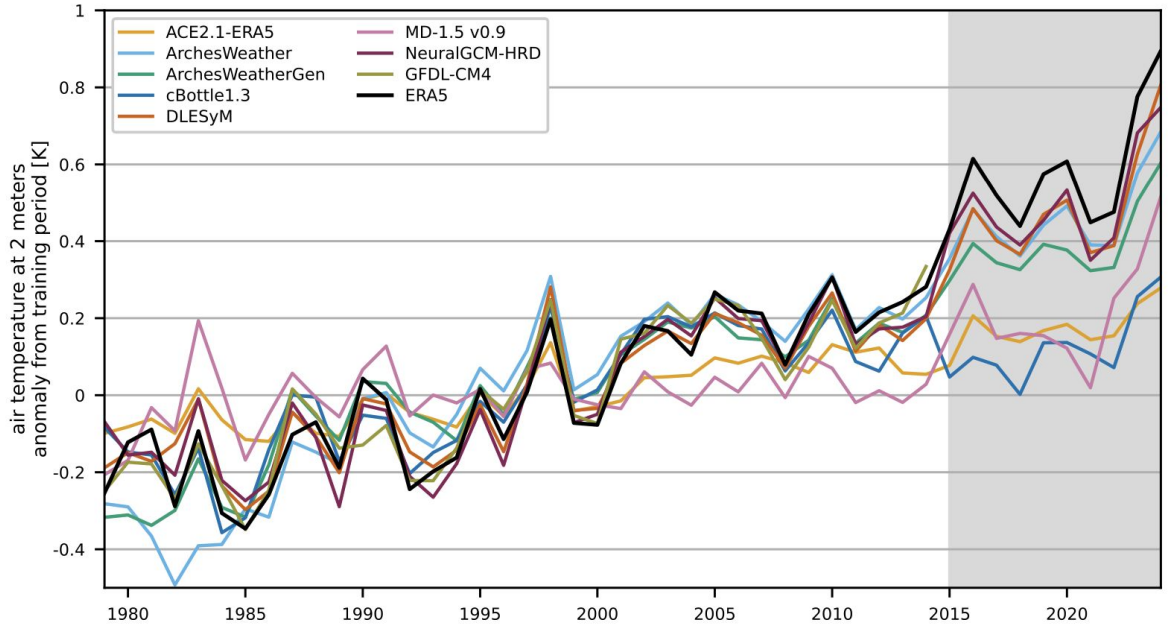
E1: global area-weighted bias pattern RMSE



RMSB area-weighted over the globe on the 1◦ grid. (a) through (g): surface variables; (h) 500 hPa geopotential height; (i) through (l), (m) through (p): temperature, specific humidity, and u, v wind at 850 hPa and 250 hPa, respectively. Bars indicate the ensemble medians and error bars indicate the ensemble ranges.

Fig. 2, Henn et al. (2026), AIMIP Phase 1: systematic evaluations of AI weather and climate models, arXiv:2605.06944.

E1: global annual average anomaly



Global- and annual-mean 2-meter air temperature, shown as anomalies from the training period (1979-2014) average. ERA5 is in black; AIWCM model ensemble means are shown, along with the CMIP6 GFDL-CM4 single-member prediction. The AIMIP test period (2015-2024) is shaded at right.

Fig. 3, Henn et al. (2026), AIMIP Phase 1: systematic evaluations of AI weather and climate models, arXiv:2605.06944.

E2: Train-period and test-period linear trends

Linear trends on annual data

Trends of global- and annual-mean variables. (a through e) surface variables, (f) 500 hPa geopotential height, (g), (h) 850 hPa temperature and humidity, and (i), (j) 250 hPa temperature and humidity. In (d) mean sea level pressure trend is shown for all models that submitted this variable, but for ACE2.1-ERA5, MD-1.5 v0.9 and NeuralGCM surface pressure trend is shown. The dark background bar is ERA5. GFDL-CM4 trends are unavailable in the test period.

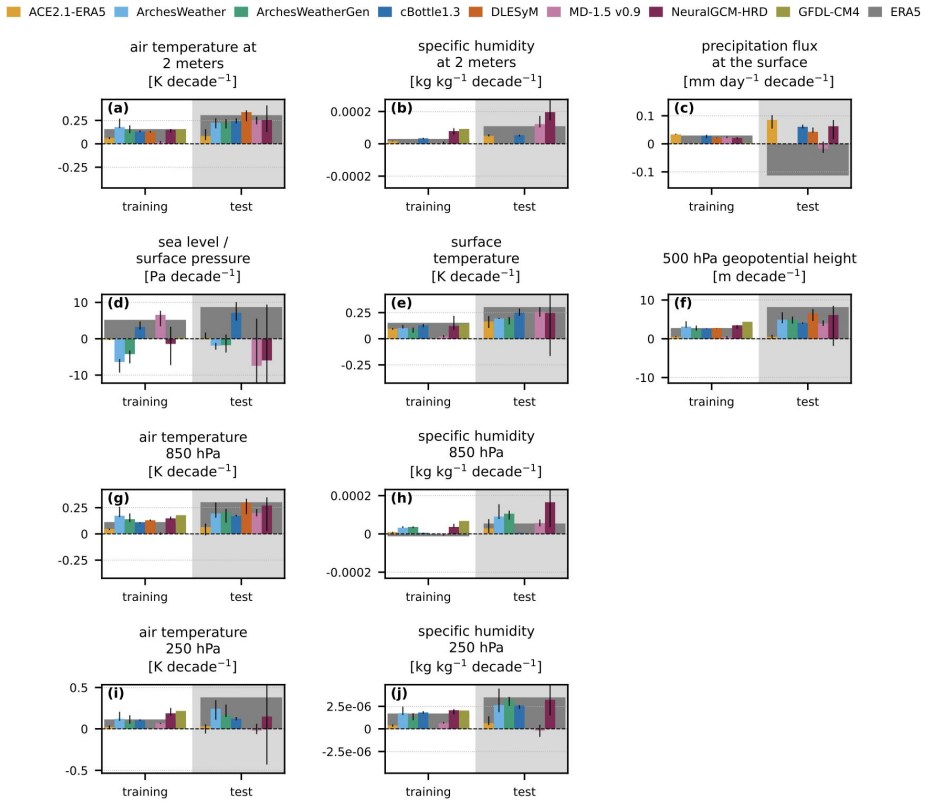


Fig. 4, Henn et al. (2026), AIMIP Phase 1: systematic evaluations of AI weather and climate models, arXiv:2605.06944.

E2: Train-period and test-period linear trends

Trend maps at 1◦ resolution over the training period for (a) 2-meter temperature and (b) surface precipitation.

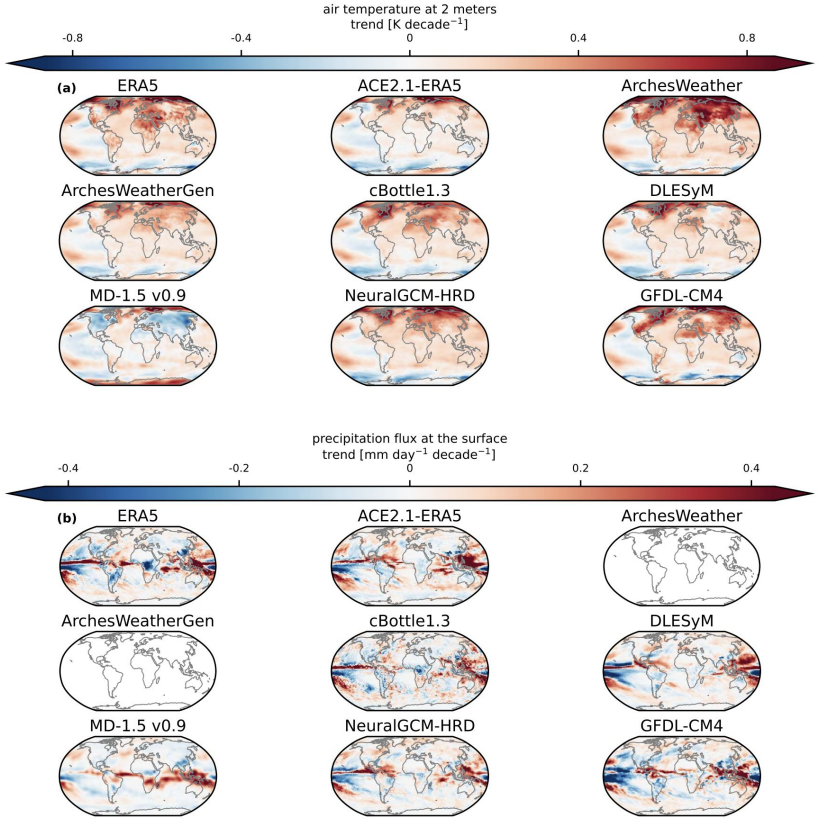
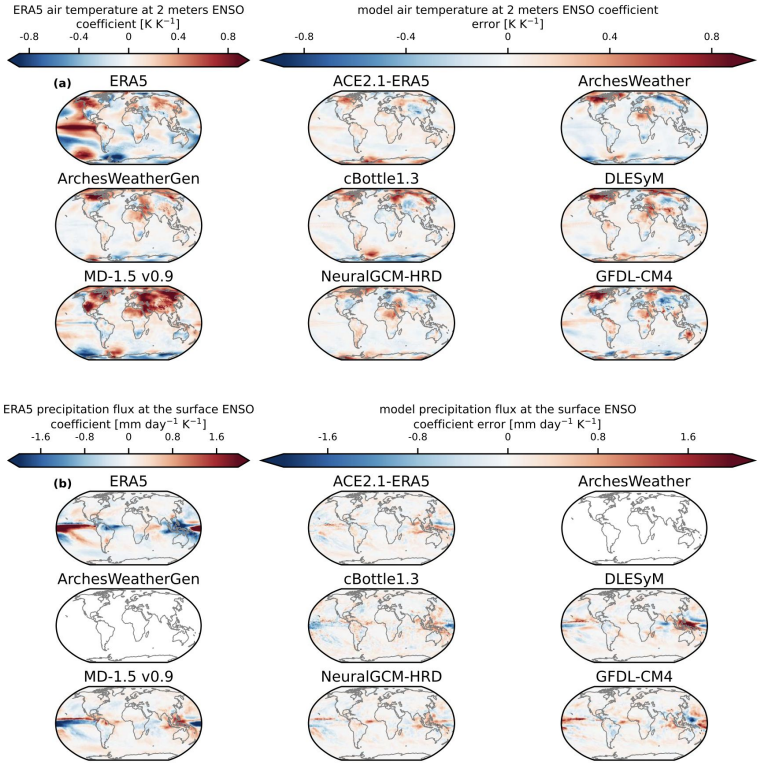


Fig. 5, Henn et al. (2026), AIMIP Phase 1: systematic evaluations of AI weather and climate models, arXiv:2605.06944.

E3: Regressions of predicted fields on the Niño3.4 index



ENSO coefficient maps at 1◦ resolution for ERA5 (upper left panels) and model coefficient errors versus ERA5 coefficients (subsequent panels) over the training period, for (a) 2-meter temperature and (b) surface precipitation.

Fig. 6, Henn et al. (2026), AIMIP Phase 1: systematic evaluations of AI weather and climate models, arXiv:2605.06944.

E4: Daily variability

Standard deviation of daily anomalies from monthly mean at 1◦ resolution over 1979, for 2-meter air temperature (a) and surface precipitation (b). Upper left panels shows anomaly standard deviation in ERA5, and subsequent panels show the error in model anomaly standard deviations.

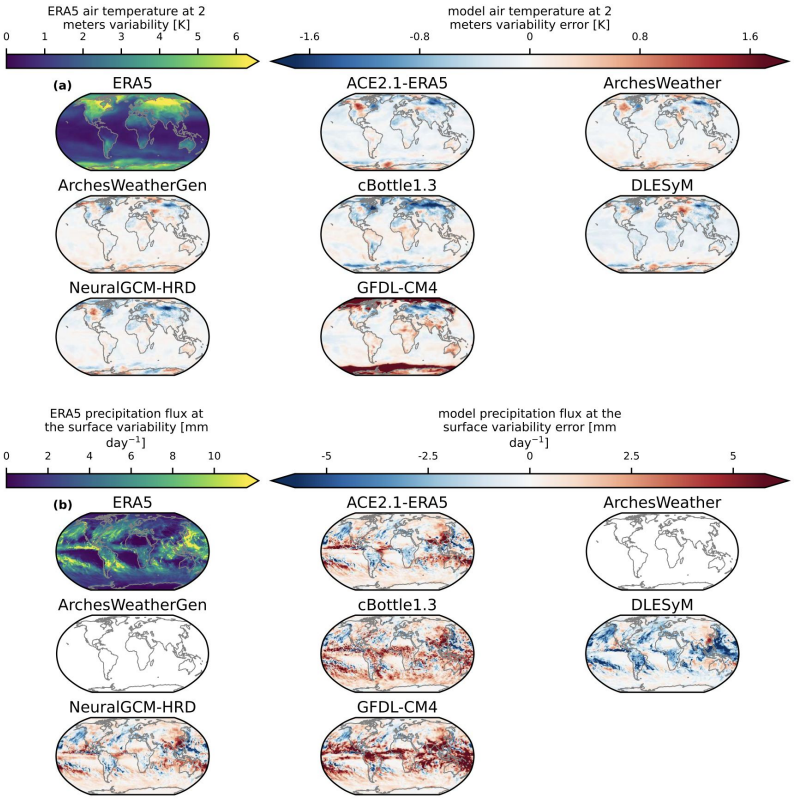


Fig. 7, Henn et al. (2026), AIMIP Phase 1: systematic evaluations of AI weather and climate models, arXiv:2605.06944.

E5: Atmospheric response to +2 K and +4 K uniform SST perturbation

Global area-weighted mean of model daily anomaly standard deviation errors, relative to global-mean ERA5 daily variability, at 1◦ resolution for the set of variables shown in Fig. 2. CMIP6 GFDL-CM4 daily variables are not available for pressure-level variables.

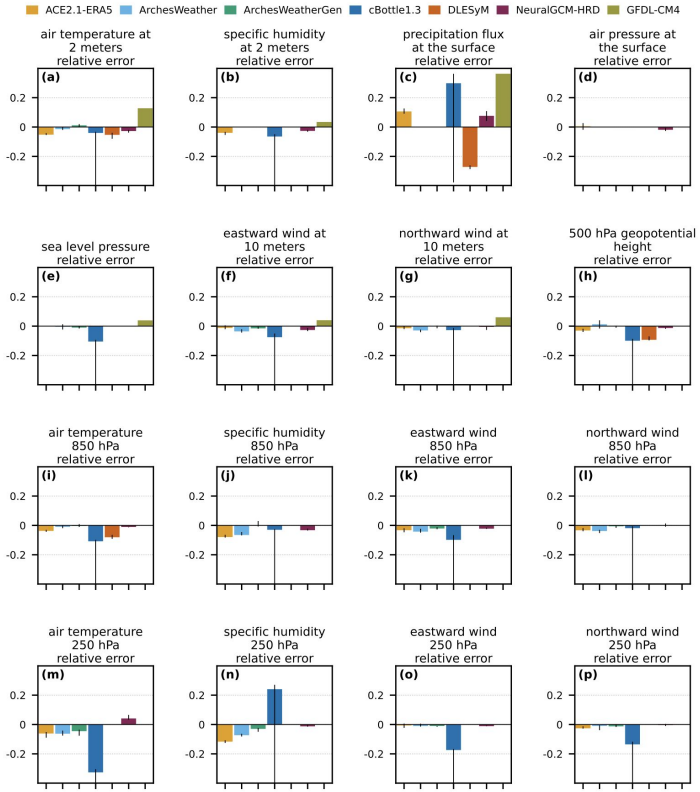


Fig. 8, Henn et al. (2026), AIMIP Phase 1: systematic evaluations of AI weather and climate models, arXiv:2605.06944.

E5: Atmospheric response to +2 K and +4 K uniform SST perturbation

Time-mean response to +2 K and +4 K SST perturbations, for 2-meter air temperature (a), (b) and surface precipitation (c), (d), respectively. Only +4 K SST perturbations are available for the GFDL-CM4 model.

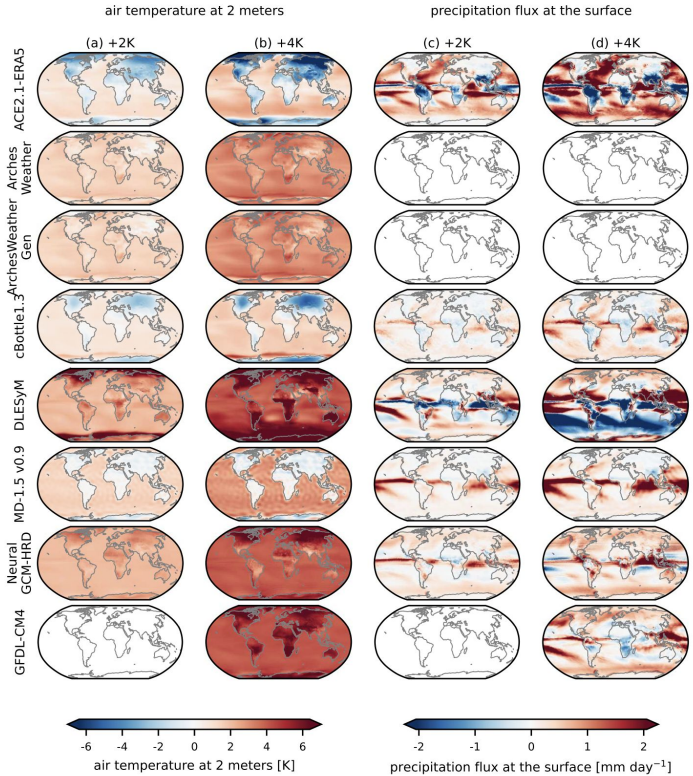


Fig. 9, Henn et al. (2026), AIMIP Phase 1: systematic evaluations of AI weather and climate models, arXiv:2605.06944.

Discussion

How well do AIWCMs do versus traditional climate models?

- Lower climate biases against ERA5 (i.e. in-distribution), versus GFDL-CM4
- Trends and out-of-sample: much harder for these models
  - Large disagreements between models, possibly due to handling of land

- Next Phase of AIMIP,
  - more complex procedures than specified-SST AMIP-type experiments (e.g., coupled earth system model, as in CMIP)
  - more extensive data outputs
  - more evaluation metrics

Next Phase of AIMIP

- more complex procedures than specified-SST AMIP-type experiments (e.g., coupled earth system model, as in CMIP)
- more extensive data outputs
- more evaluation metrics

What are your ideas?

Questions?

Citations

Henn, B., Bretherton, C. S., Koldunov, N., Lessig, C., Molina, M. J., Arcomano, T., Watt-Meyer, O., Couairon, G., Singh, R., Brunstein, R., Hasson, Y., Jost, A., Brenowitz, N., Manshausen, P., Cresswell-Clay, N., Durran, D., Hall, K. J. C., Yuval, J., Kochkov, D., Hoyer, S., & Lopez-Gomez, I. (2026). *AIMIP Phase 1: systematic evaluations of AI weather and climate models.* arXiv:2605.06944 https://doi.org/10.48550/arXiv.2605.06944